# Background

Infertility affects approximately 186 million people worldwide and 8-12% of couples of reproductive age. Therefore, a comprehensive diagnostic evaluation of infertility is crucial to achieving improvements in targeted prevention and treatment outcomes (Wasilewksi, 2020).

## Poor Prognosis and Futility

Examples where continued treatment may be futile: (ASRM, 2019)

FSH level ≥ 15mIU/mL; OR AMH level < 0.2 ng/mL (LaMarca, 2013); OR Antral follicle count < 3 (ASRM, 2021(a); LaMarca, 2013). Lack of viable spermatozoa. Ovarian failure where a couple is attempting conception with their own gametes. Numerous ART cycles without adequate egg production, fertilization and/or embryo development.

National data from the SART registry 2019 demonstrates that the cumulative live birth per intended retrieval resulting in live births decreased progressively from: 55.0% in females younger than 35 years; 41.0% for females aged 35-37 years; 26.8% for females aged 38-40 years; 13.4% for females aged 41-42; and 4.14% for females over the age of 42. The age-related decline in fertility is accompanied by a significant increase in the rates of aneuploidy and spontaneous abortion. (SART, 2020)

***Poor Prognosis:*** "Very poor prognosis'' refers to treatment for which the odds of achieving a live birth are very low but not nonexistent (>1% to <5% per cycle). (ASRM, 2019)

## Aneuploidy

Female fecundity decreases with increasing age. Aging chromosome segregation errors during meiotic division are increasingly common and lead to the production of oocytes an incorrect number of chromosomes, referred aneuploidy. trophectoderm biopsies >15,000 blastocysts have shown that the rate of aneuploidy steadily increases after age 31 and reaches 85% at age 43 (Franasiak, 2014).

To determine an 85% risk for aneuploidy, a healthcare professional typically uses a prenatal screening test, like a combined first-trimester screening (cFTS), which combines maternal blood tests measuring specific markers (like PAPP-A and free β-hCG) with an ultrasound measurement of the fetal nuchal translucency (NT), and interprets the results based on the mother's age to calculate a risk score; a score indicating an 85% chance of aneuploidy would be considered "high risk" and usually warrants further diagnostic testing like chorionic villus sampling (CVS) or amniocentesis to confirm the diagnosis

The risk calculation considers maternal age, ultrasound findings (like NT thickness), and levels of specific biochemical markers in the blood (Russo, 2014) (Harris et.al., 2018).